# Data_Scientist — Step 2

## Master Statistics & Mathematics

Detailed Learning Notes • CareerCompass

## 1. Learning Objective

Develop the quantitative reasoning needed for statistical analysis, experimentation and machine-learning interpretation.

## 2. Core Concepts — Detailed Breakdown

### Descriptive statistics

Use distributions, central tendency, spread and quantiles to understand data before modeling.

### Probability

Understand events, conditional probability, independence, random variables and expectation.

### Inference

Study sampling, confidence intervals, hypothesis testing, effect sizes and statistical power conceptually.

### Linear algebra

Know vectors, matrices, matrix multiplication and the geometry of projections because many ML algorithms operate on matrix representations.

### Optimization

Understand derivatives, gradients and the idea of minimizing an objective function.

## 3. Recommended Learning Workflow

- Complete statistics exercises.
- Simulate probability experiments in Python.
- Practice confidence intervals and hypothesis tests.
- Review vectors/matrices and simple optimization.
- Apply the concepts to a real dataset.

## 5. Hands-on Project / Practice

Run a controlled simulation comparing two groups. Generate samples, estimate means, visualize sampling variability, compute confidence intervals, perform a suitable test, calculate an effect size, and explain why the result may or may not generalize.

## 6. Common Mistakes & Pitfalls

- Reporting p-values without effect size.
- Ignoring assumptions behind a statistical method.
- Treating sample estimates as exact population values.
- Using complex math without connecting it to the data problem.

- Confusing prediction with causal inference.

## 7. Completion Checklist

- I can explain sampling variability.
- I can interpret confidence intervals.
- I can explain hypothesis testing and effect size.
- I can use probability to reason about uncertainty.
- I understand the role of linear algebra and optimization in ML.

## 8. Interview / Assessment Questions

- What is a confidence interval?
- What is statistical power?
- Why is effect size important?
- What is conditional probability?
- Where does linear algebra appear in ML?

## 9. Recommended References

Khan Academy Statistics & Probability: https://www.khanacademy.org/statistics-probability

Data Science Tutorial: https://www.youtube.com/watch?v=ua-CiDNNj30

Machine Learning in 2024: https://www.youtube.com/watch?v=bmmQA8A-yUA

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.